LS2101 Endsem Exam, Autumn 2025

06.12.2025

Part -A, Total Marks 30

Answer all questions (*write the answers for Part A in the main answer script*)

1. Why physiological buffers are important? Give two examples of physiological buffers. (1+2 = 3)

2. Which is the first committed step of glycolysis? Briefly explain your answer. (1+2 = 3)

3. Using your knowledge of energy metabolism predict the intracellular localization of the following enzymes or transporters. Provide a 'one line' explanation for each: a) glucose transporter (GLUT4); b) pyruvate kinase; c) pyruvate carrier; d) pyruvate dehydrogenase. (1.5 X 4 = 6)

4. How does the cell ensure that glycogen synthase (building glycogen) and glycogen phosphorylase (breaking glycogen) are not active at the same time? (3)

5. Why does the cell quickly convert glucose to glucose-6-phosphate as soon as it enters? Which enzyme catalyses this reaction? (2+1 = 3)

6. A metabolic pathway converts substrate A into product X through several intermediate steps. Explain how feedback inhibition prevents the unnecessary accumulation of the final product X. (2)

7. How premature activation of digestive enzymes is prevented in the pancreas and stomach? Why such regulation is important? (2+1 = 3)

8. What are pH optima of enzymes? Logically explain what should be the pH optima of a lysosomal hydrolase and a cytoplasmic kinase? (1+2 = 3)

9. What is the ultimate source of carbon in living system? Explain using a flow chart how biospheric carbon is assimilated by animals. (1+3 = 4)